IF

What does it do?

It checks whether a condition is true or not and on the basis of that returns a value

Syntax

=if(Logical_Test, [value_if_true], [value_if_false])

Logical_Test

This will be the condition which you want to check, so greater than, less than or equal to signs can be used for evaluate a condition based on text value, then use double quotes ('') with equal to signs (Row 27 for example)

Value_if_True

If the logical test holds, then excel will return this value

Value_if_False

If the logical test does not hold, then excel will return this value

Example

Using IF formula to identify if a student has passed an exam or not

Student ID	Marks	Pass or Fail	Formula
S001	90	**Pass**	=IF(B22>=40,"Pass","Fail")
S002	70	**Pass**	=IF(B23>=40,"Pass","Fail")
S003	20	**Fail**	=IF(B24>=40,"Pass","Fail")
S004	41	**Pass**	=IF(B25>=40,"Pass","Fail")

Note

If instead of marks, we had Grades A, B, C i.e. text values then the above formulas would have changed by IF(B21= 'A',"Pass","Fail")

Exercise

Write a formula using IF. Salary for an AVP will be $2,000 and for an analyst will be $1,000

Emp ID	Designation	Salary
E001	AVP	**$2,000**
E002	Analyst	**$1,000**
E003	Analyst	**$1,000**
E004	Analyst	**$1,000**

numbers. When you want to

AND

What does it do?

Checks if all arguments are true and returns TRUE value else returns FALSE value. You need to have minimu
function is used with IF condition a lot

Syntax

=AND(Logical 1, Logical 2, Logical 3, and so on..)

Logical 1

First condition which needs to be checked

Logical 2

Second condition which needs to be checked

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using AND

Condition 1	Condition 2	Result	Formula
1	2	TRUE	=AND(A19>0,B19>0)
1	-2	FALSE	=AND(A20>0,B20>0)

Example 2

Use IF with AND function to check if student has passed or not. If the student has failed in 1 of the 2 subjec
considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	PASS	=IF(AND(B26="PASS",C26="PASS"),'
S002	PASS	FAIL	FAIL	=IF(AND(B27="PASS",C27="PASS"),'
S003	FAIL	PASS	FAIL	=IF(AND(B28="PASS",C28="PASS"),'
S004	PASS	PASS	PASS	=IF(AND(B29="PASS",C29="PASS"),'

Exercise

Write a formula using AND with IF to find if an employee should get a "GOOD" increment or not. Sales mac
more than 50 and he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)
E001	AVP	**51**	**YES**	YES
E002	Analyst	**100**	**NO**	NO
E003	Analyst	**49**	**YES**	NO
E004	Analyst	**75**	**YES**	YES

ım 2 conditions to check. AND

:ts, then the student will be

"PASS","FAIL")
"PASS","FAIL")
"PASS","FAIL")
"PASS","FAIL")

de by the employee should be

Nested IF's

What does it do?

Nested IF conditions help in checking multiple conditions together by using IF conditions within IF condition.

Syntax

=if(Logical_Test, [value_if_true], if(Logical_Test, [value_if_true], [value_if_false]))

Logical_Test

This will be the condition which you want to check, so greater than, less than or equal to signs can be used f
condition based on text value, then use double quotes ('') with equal to signs

Value_if_True

If the logical test holds, then excel will return this value

Value_if_False

If the logical test does not hold, then excel will return this value

Nested IF

From Excel 2007 onwards, we can use 64 IF conditions in one formula whereas Excel 2003 allows for ony 7 IF

Example

Use Nested IF to put the grade a student gets in an exam in Table B. Score Ranges for different grades are gi

TABLE A	
Score	**Grade**
> 70	A
41-70	B
< 40	C

TABLE B			
Student ID	**Marks**	**Grade**	**Formula**
S001	90	A	=IF(E26< 41,"C",IF(E26
S002	70	B	=IF(E27< 41,"C",IF(E27
S003	20	C	=IF(E28< 41,"C",IF(E28
S004	41	B	=IF(E29< 41,"C",IF(E29

Exercise

Write a formula using Nested IF conditions to check what level increment an employee should get and then
dollars using Format

TABLE A	
Performance	**Increment**
A	$1,000
B	$500
C	$250

TABLE B			
Emp ID	**Designation**	**Performance**	**Result**
E001	AVP	**A**	1000
E002	Analyst	**A**	1000
E003	Analyst	**B**	500
E004	Analyst	**C**	250

or numbers. When you want to evaluate a

conditions

ven in Table A.

< 71,"B","A"))
< 71,"B","A"))
< 71,"B","A"))
< 71,"B","A"))

convert the amount into

OR

What does it do?

Checks if any of the conditions are true or not. It returns TRUE value even if one of the conditions holds els
a FALSE to appear, no conditions should hold. You need to have minimum 2 conditions to check. AND funct

Syntax

=OR(Logical 1, Logical 2, Logical 3, and so on..)

Logical 1

First condition which needs to be checked

Logical 2

Second condition which needs to be checked

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using OR

Condition 1	Condition 2	Result	Formula
1	2	TRUE	=OR(A20>0,B20>0)
1	-2	TRUE	=OR(A21>0,B21>0)
-1	-2	FALSE	=OR(A22>0,B22>0)

Example 2

Use IF with OR function to check if student has passed or not. If the student has failed in 1 of the 2 subjects
considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	PASS	=IF(OR(B28="FAIL",C28="FAIL"),"F
S002	PASS	FAIL	FAIL	=IF(OR(B29="FAIL",C29="FAIL"),"F
S003	FAIL	PASS	FAIL	=IF(OR(B30="FAIL",C30="FAIL"),"F
S004	PASS	PASS	PASS	=IF(OR(B31="FAIL",C31="FAIL"),"F

Exercise

Write a formula using OR with IF to find if an employee should get a "GOOD" increment or not. Either Sales
more than 50 or he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)
E001	AVP	**51**	**YES**	YES
E002	Analyst	**49**	**NO**	NO
E003	Analyst	**49**	**YES**	YES
E004	Analyst	**75**	**YES**	YES

e returns FALSE value. Basically, for
tion is used with IF condition a lot

s, then the student will be

FAIL","PASS")
FAIL","PASS")
FAIL","PASS")
FAIL","PASS")

s made by the employee should be

IFERROR

What does it do?

Sometimes when we use formula in cells, we might get an error because of the cell value, IFERROR allows u
we want

Syntax

=IFERROR(Value,[Value_if_error])

Logical 1

First cell value which is checked if there is an error

Value_if_error

If in the cell there is an error, that error value will be replaced by this value

Example 1

Divide Col A by Col B and show the results without using IFERROR and with using IFERROR

Col A	Col B	Without IFERROR &Formula		With IFERROR &Formula	
1	2	0.5	=A19/B19	0.5	=IFERROR(A19/B19,999
1	0	**#DIV/0!**	=A20/B20	**999**	=IFERROR(A20/B20,999

Note

In the above example, on dividing 1 by 0, we get an error, without using IFERROR we get #DIV/0! i.e. an err
values we want. You can also enter text which can let the user know that there is some problem with the d

Example 2

Divide Col A by Col B and show the results without using IFERROR and with using IFERROR

Col A	Col B	Without IFERROR &Formula		With IFERROR &Formu	
1	2	2	=A29*B29	2	=IFERROR(A29*B29,
1	1	1	=A30*B30	1	=IFERROR(A30*B30,
1	a	**#VALUE!**	=A31*B31	**DATA IS WRONG**	=IFERROR(A31*B31,
5	a	**#VALUE!**	=A32*B32	**DATA IS WRONG**	=IFERROR(A32*B32,

Exercise

Write a formula using IFERROR. Find the monthly salary of each employee using data in Column C and if an
ERROR IN SALARY. Change the format to £ (Pounds) and no decimal places

Emp ID	Designation	Annual Salary	Monthly Salary
E001	AVP	**£80,000**	**6667**
E002	Analyst	**£50,000**	**4167**
E003	Analyst	**abcd**	**ERROR IN SALARY**
E004	Analyst	**£40,000**	**3333**

ıs to remove such errors with values

)
)

or. So using IFERROR, we can give
ata.

ıla
"DATA IS WRONG")
"DATA IS WRONG")
"DATA IS WRONG")
"DATA IS WRONG")

The red color highlight is for representation here - color cannot be changed using IFERROR

error appears, then change it to

IFNA

What does it do?

Sometimes when we use formula in cells, we get values errors of type #N/A. We can use IFNA to remove th
function is from Excel 2013 onwards, so is not available in earlier versions of Excel.

Syntax

=IFNA(Value,[Value_if_error])

Logical 1

First cell value which is checked if there is an error

Value_if_error

If in the cell there is a value of #N/A, the error value will be replaced by this value

Example 1

Divide Col A by Col B and show the results without using IFERROR and with using IFERROR

Col A	With IFERROR &Formula	
1	1	=IFNA(A19,**9999**)
#N/A	**9999**	=IFNA(A20,**9999**)

**The red color highlig
representation here - c
be changed using**

Exercise

Write a formula using IFNA. Find the monthly salary of each employee using data in Column C and if #N/A a
SALARY. Change the format to £ (Pounds) and no decimal places

Emp ID	Designation	Annual Salary	Monthly Salary
E001	AVP	**£80,000**	**6667**
E002	Analyst	**£50,000**	**4167**
E003	Analyst	**#N/A**	**#N/A**
E004	Analyst	**£40,000**	**3333**

ıis with our choice of value. IFNA

ght is for
olor cannot
IFNA

appears, then change it to ERROR IN

EXACT

What does it do?

It checks whether 2 values (texts, numbers) are exactly same or not. If they are same, then it will return TR
function is case-sensitive

Syntax

=EXACT(Text1, Text2)

Text1

First string which you want to check

Text2

Second string which you want to test against

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using AND

Col A	Col B	Result	Formula
EXCEL	EXCEL	TRUE	=EXACT(A19,B19)
EXCEL	Excel	FALSE	=EXACT(A20,B20)

Note

In the above example, Row 20 result is FALSE because the EXACT function is case-sensitive. The first value i
the second one (Excel) is a combination of upper and lower capse

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Col A	Col B	Results
TEST	Test	**FALSE**
TEST	TEST	**TRUE**
1234	1234	**TRUE**
1.1	1.101	**FALSE**

:UE else it will return FALSE. This

s in upper capse (EXCEL) whereas

TEXT

What does it do?

It converst a numeric value into text based on the format given by the user

Syntax

=TEXT(Value, Format_text)

Value

Value for which we need to change the format

Format_text

Format specified by the user

Example 1

Using different text formats

Col A	Result	Formula
27/03/2017	27/03/2017	=TEXT(A18, "dd/mm/yyyy")
42821	27/3/2017	=TEXT(A19, "dd/m/yyyy")
42821	Mar 27 2017	=TEXT(A20, "mmm dd yyyy")
27/03/2017	Mar-17	=TEXT(A21,"mmm-yy")
14.141414	14.14	=TEXT(A22, "0.00")
14598.12	$14,598.12	=TEXT(A23, "$#,##0.00")
14598.12	£14,598.12	=TEXT(A24, "£#,##0.00")
14:00	14:00	=TEXT(A25, "hh:mm")

Using Tex
appear in t

Note:

Col A Row 19 & 20 - Excel stores dates in numbers, so that is why when we use Text with Date formats, Exce

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Value	Convert To	Results
21/05/2017	21/05/2017	**21/5/2017**
42876	21/05/2017	**21/5/2017**
12:00	12:00	**00/1/1900**
12345.78	£12,345.78	**18/10/1933**
12345.78	£12,346	**18/10/1933**

:t function though the results may
he same way, but there format has
changed to text.

l converts those numbers dates

CONCATENATE

What does it do?

This function joins several (more than 1) text string into 1 string

Syntax

=CONCATENATE(String1, String2, and so on..)

OR

=String1&String2& so on..

String 1

First string which you want to concatenate

String 2

Second string which you want to join to the first string

Example

Join the 3 strings together using CONCATENATE and & (ampersand) operator

String1	String2	String3	Result	Formula
S001	90	Pass	**S00190Pass**	=CONCATENATE(A20,B20,C20)
S002	70	Pass	**S00270Pass**	=CONCATENATE(A21,B21,C21)
S003	20	Fail	**S00320Fail**	=A22&B22&C22
S004	41	Pass	**S00441Pass**	=A23&B23&C23
S005	39	Fail	**S005 39 Fail**	=CONCATENATE(A24," ",B24," ",C24)
S006	99	Pass	**S006 99 Pass**	=A25&" "&B25&" "&C25
S007	11	Fail	**S007;11;Fail**	=CONCATENATE(A26,";",B26,";",C26)

Note

In order to add spaces, other special characters between strings, you would need to use double quotes "" a
done in the examples above

Exercise

Join the 3 strings together using CONCATENATE and & (ampersand) operator to get the results as mentione

String1	String2	String3	Result	USE	Formula
E001	AVP	800	**E001-AVP-800**	CONCATENATE	**E001-AVP-800**
E002	Analyst	500	**E002Analyst500**	CONCATENATE	**E002Analyst500**
E003	Analyst	100	**E003/Analyst-100**	CONCATENATE	**E003/Analyst-10(**
E004	Analyst	400	**E004:Analyst:400**	&	**E004:Analyst:400**
E005	AVP	1000	**E005 AVP 1000**	&	**E005:AVP:1000**
E006	Analyst	250	**E006 Analyst 250**	&	**E006 Analyst 250**
E007	Analyst	500	**E007;Analyst*500**	&	**E007;Analyst*50(**

In these 3 examples, we have added spaces or semi-colons between the strings to make the result readable

and declare the value between them as

ed in the result column

TRIM

What does it do?

The function removes spaces all before and following the words i.e. leading and trailing spaces - and also r
but does not remove the single space between words. TRIM function is used to clean data.

Syntax

=TRIM(Text)

Text

First string which you want to concatenate

Example

Use TRIM to remove spaces

String1	Result	Formula
S001 A	**S001 A**	=TRIM(A16)
S001 A	**S001 A**	=TRIM(A17)
S002	**S002**	=TRIM(A18)
S003	**S003**	=TRIM(A19)

Exercise

Remove the extra spaces in the strings and then join the 3 strings together using CONCATENATE and & (am
mentioned in the result column. Do so in one formula itself

String1	String2	String3	Result	USE	Formula
E001	AVP	800	**E001-AVP-800**	CONCATENATE	**E001-AVP-800**
E002	Analyst	500	**E002Analyst500**	CONCATENATE	**E002Analyst500**
E003	Ana lyst	100	**E003/Analyst-100**	&	**E003/Ana lyst-10(**
E004	Analyst	400	**E004:Analyst:400**	&	**E004:Analyst:400**

emoves extra spaces between words

ipersand) operator to get the results as

)
)

SUBSTITUTE

What does it do?

This function is used to replace a part of text in a string with something else. SUBSTITUTE function is used t
sensitive.

Syntax

=SUBSTITUTE(Text, Old_Text, New_Text, [instance_number])

Text

String in which you want to modify text

Old_Text

The old text which you want to replace

New Text

The new text with which you want to modify your original text with

Instance_Number

The instance of old text which you want to replace with new text. It is optional but if not provided it will m
old text with new text. Look at the examples below for better understanding of this

Example

Join the 3 strings together using CONCATENATE and & (ampersand) operator

String	Instances	Result	F
Hut		Hub	=SUBSTIT
bumble	1	Tumble	=SUBSTITU
the car is in the driveway		car is in driveway	=SUBSTITU
the car is in the driveway	2	the car is in a driveway	=SUBSTITU
The car is in the driveway		car is in the driveway	=SUBSTITU

Note

If instances are not mentioned, then excel will replace all the occurrences as can be seen on Row 28. And t
be seen on Row 30

Exercise

Use SUBSTITUTE function to make changes to some of the strings below to get the results mentioned in th

String	Result	Formula
The dog is in the lawn	dog is in the lawn	**dog is in the lawn**
the dog is in the lawn	Dog is in lawn	**dog is in lawn**
Dog	Fog	**Fog**
Curious Cat	Furious Cat	**Furious Cat**

:o clean data. This function is case-

ake changes to all the occurrences of

Formula
UTE(A26,"t","b")
JTE(A27,"b","T",1)
JTE(A28,"the"," ")
TE(A29,"the","a",2)
JTE(A30,"The"," ")

he function is case-sensitive which can

e 'Result' column

VALUE

What does it do?

The function is used to convert text value into numeric value

Syntax

=VALUE(Text)

Text

String which you want to convert to number

Example

Convert the values into numbers

Text	Result	Formula
$12,345	**12345**	=VALUE(A15)
27/03/2017	**42821**	=VALUE(A16)
09/12/2014	**41982**	=VALUE(A17)

Excel stores dates as numbers, hence t changes the dates to numbe

Exercise

Convert the values in Text column into numbers

Text	Result	Formula
$123.17	**123**	**123**
01/12/2017	**43070**	**43070**
12/01/2017	**42747**	**42747**

he function
rs

UPPER

What does it do?

This function converts the entire text string into uppercase letters

Syntax

=UPPER(Text)

Text

String for which you want to convert all letters into UPCASE

Example

Convert the values in Text column into UPPERCASE

Text	Result	Formula
Laptop	LAPTOP	=UPPER(A15)
excel	EXCEL	=UPPER(A16)
superstore is closed now	SUPERSTORE IS CLOSED NOW	=UPPER(A17)

Exercise 1

Convert the values in Text column into UPPERCASE

Text	Formula
good	**GOOD**
how are you?	**HOW ARE YOU?**

Exercise 2

Use UPPER to change Col A and then use EXACT function to match the 2 strings to see if they are same or not

Col A	Col B	Result
match	MATCH	MATCH
match	Match	MATCH
correct	CORECT	CORRECT

LOWER

What does it do?

This function converts the entire text string into lowercase letters

Syntax

=LOWER(Text)

Text

String for which you want to convert all letters into lowercase

Example

Convert the values in Text column into lowercase

Text	Result	Formula
Laptop	laptop	=LOWER(A15)
EXCEL	excel	=LOWER(A16)
SUPERSTORE IS CLOSED NOW	superstore is closed now	=LOWER(A17)

Exercise 1

Convert the values in Text column into lowercase

Text	Formula
GOOD	**good**
HOW are you?	**how are you?**

Exercise 2

Use LOWER to change Col A and then use EXACT function to match the 2 strings to see if they are same or

Col A	Col B	Result
MATCH	match	match
MATCH	mATCH	match
CORECT	correct	corect

not

LEN

What does it do?

This function returns the number of characters in a string (text or numeric value). It also counts the space is used for data manipulation along with LEFT, RIGHT functions

Syntax

=LEN(Text)

Text

String for which you want to find the length

Example

Convert the values in Text column into lowercase

Text	Result	Formula
Laptop	6	=LEN(A16)
EXCEL	6	=LEN(A17)
SUPERSTORE IS CLOSED ;	22	=LEN(A18)
90	2	=LEN(A19)

Exercise 1

Find the length of the followig strings

Text	Formula
Phone Number is +(0)1234	**24**
This is used for data cleaning	**30**

s and special characters. This function

LEFT

What does it do?

This function returns the number of specified characters from the start of the string. This function is used for da
from left to right - so that is why from the start of the string

Syntax

=LEFT(Text, [num_chars])

Text

String from which you want to find letters

num_chars

Number of characters you want to read from the left

Example

Use LEFT function on the following strings

Text	Result	Formula
Laptop	Lapto	=LEFT(A19,5)
EXCEL IS WOW	EXCEL IS	=LEFT(A20,8)
SUPERSTORE IS CLOSED ;	SUPERSTORE IS C	=LEFT(A21,15)
90	9	=LEFT(A22,1)
SUPERSTORE IS CLOSED ;	SUPERSTORE IS C	=LEFT(A23,LEN(A23)-7)

Exercise 1

Use LEFT function to get the results in the 2nd column

Text	Result	Formula
Phone Number is +(0)1234	Phone Number is +(0	**Phone Number**
This is used for data cleaning	This is used for data cle	**This is used fo**

Exercise 2

Use LEFT and LEN function together to only remove last 5 characters of the string

Text	Result	Formula
Phone Number is +(0)1234	Phone Number is +(0	**Phone Number**
This is used for data cleaning	This is used for data cle	**This is used for c**

ta manipulation. A way to remember is that you read

is +(0)
r dat

is +(0)
data cle

RIGHT

What does it do?

This function returns the number of specified characters from the end of the string. This function is used for data that you read from left to right - so that is why from the end of the string

Syntax

=RIGHT(Text, [num_chars])

Text

String from which you want to find letters

num_chars

Number of characters you want to read from the left

Example

Use RIGHT function on the following strings

Text	Result	Formula
Laptop	top	=RIGHT(A19,3)
EXCEL IS WOW	WOW	=RIGHT(A20,4)
SUPERSTORE IS CLOSED ;	OSED ;	=RIGHT(A21,6)
90	0	=RIGHT(A22,1)
SUPERSTORE IS CLOSED ;	OSED ;	=RIGHT(A23,LEN(A23)-16)

Exercise 1

Use RIGHT function to get the results in the 2nd column

Text	Result	Formula
Phone Number is +(0)1234	is +(0)1234	**is +(0)1234**
This is used for data cleaning	data cleaning	**data cleaning**

Exercise 2

Use RIGHT and LEN function together to only remove first 5 characters of the string

Text	Result	Formula
Phone Number is +(0)1234	Number is +(0)1234	**Number is +(0)1234**
This is used for data cleaning	is used for data cleaning	**is used for data cleaning**

a manipulation. A way to remember is

MID

What does it do?

This function returns the number of specified characters from the left to right but, you can change the sta
function different from LEFT function.

Syntax

=MID(Text, [starting_position], [num_chars])

Text

String from which you want to find letters

starting_position

Starting position of the string

num_chars

Number of characters you want to read from the starting of the string

Example

Use RIGHT function on the following strings

Text	Result	Formula
Laptop restarted	top re	=MID(A22,4,6)
EXCEL IS WOW	XCEL IS W	=MID(A23,2,9)
SUPERSTORE IS CLOSED ;	E IS CLOS	=MID(A24,10,9)
90 EMPLOYEES	0 EMPL	=MID(A25,2,6)

Exercise

Use MID function to get the results in the 2nd column

Text	Result	Formula
Phone Number is +(0)1234	is +(0)1	**is +(0)1**
This is used for data cleaning	for data clean	**for data clean**
How have you been?	you	**you**
Any questions?	quest	**quest**

ırt of the string thus making MID

DATE

What does it do?

This function returns the date when days, month, year are stored in different cells. If the cell in which the fur
then the function will return a number - excel stores dates as numbers

Syntax

=DATE(Year,Month,Day)

Year

Year of the date. Try to supply the entire 4 digits

Month

Month of the date in numeric format like for January it would be 1, for February it would be 2

Day

Day of the date - it needs to be in numeric format like 1, 2, 3 …and till 31

Example

Use DATE function to calculate dates

Year	Month	Day	Result	Formula
2017	12	31	31/12/2017	=DATE(A22,B22,C22)
2016	11	30	30/11/2016	=DATE(A23,B23,C23)
2015	3	1	01/03/2015	=DATE(A24,B24,C24)
2014	8	1	01/08/2014	=DATE(A25,B25,C25)

Exercise 1

Use DATE function to calculate dates and then format the result into Date format

Year	Month	Day	Formula
2012	7	31	31/07/2012
2011	1	30	30/01/2011
2010	2	1	01/02/2010
2000	5	25	25/05/2000

Exercise 2

Use LEFT, MID and RIGHT functions to extract day, month and year from Col A and then use DATE function to

Col A	Year	Month	Day	Date
21122016	2016	12	21	21/12/2016
13122000	2000	12	13	13/12/2000
15011999	1999	01	15	15/01/1999
10102010	2010	10	10	10/10/2010

ıction is being called is of Number format,

ɔ calculate dates

DATEVALUE

What does it do?

This function converts a date which has been entered as text to date value but in number format

Syntax

=DATEVALUE(DATE_TEXT)

DATE_TEXT

Date which needs to be converted from text to date format

Example

Use DATE function to calculate dates

Text Date	Result	Formula
05/05/2017	42860	=DATEVALUE(A15)
02/12/2018	43436	=DATEVALUE(A16)
01/03/2015	42064	=DATEVALUE(A17)
02/08/2014	41853	=DATEVALUE(A18)

Exercise

Use DATEVALUE function to calculate dates and then change the format of the result into Date format

Text Date	Result
31/12/2015	42369
3/01/2010	40181
11/12/2015	42349
01/01/2014	41640

TODAY

What does it do?

This function returns today's date in date format

Syntax

=TODAY()

No paramter

No paramter is required for this function

Example

Identify today's date

Result	Formula
13/03/2026	=TODAY()
08/03/2026	=TODAY()-5

Exercise

Calculate number of days between today's date and date in Col A

Col A	Result
31/12/2015	3725
03/01/2010	5913
11/12/2015	3745
01/01/2014	4454

NOW

What does it do?

This function returns current date and time in excel date and time format

Syntax

=NOW()

No paramter

No paramter is required for this function

Example

Identify today's date and time

Result	Formula
13/03/2026 11:59	=NOW()
08/03/2026 11:59	=NOW()-5

Exercise

Use NOW() to calculate values in COL A

Col A	Formula
Current date and time	13/03/2026 11:59
Date and Time 7 days ago	06/03/2026 11:59

DAY

What does it do?

This function returns the day of the month i.e. 1, 2 ..31 in numeric format. Do note that at times, you may
numeric to get the date in numeric format

Syntax

=DAY(SERIAL_NUMBER)

Serial Number

Date from which you want to extract the date part

Example

Identify today's date and time

Col A	Result	Formula
13/03/2026 11:59	13	=DAY(A16)
08/03/2026 11:59	8	=DAY(A17)

Exercise

Use NOW() to calculate values in COL A and then use DAY function to extract date part

Col A	Results from NOW function	Results from D
Current date and time	13/03/2026	13
Date and Time 7 days ago	06/03/2026 11:59	6

/ have to change the format of the cell to

ate function

MONTH

What does it do?

This function returns the month in a date in a number format i.e. 1, 2, 3 ..12. Do note that at times, you m
to numeric to get the date in numeric format

Syntax

=MONTH(SERIAL_NUMBER)

Serial Number

Date from which you want to extract the month part

Example

Identify today's date and time

Col A	Result	Formula
29/03/2017 12:11	3	=MONTH(A16)
22/02/2017 12:11	2	=MONTH(A17)

Exercise

Use NOW() to calculate values in COL A and then use MONTH function to extract month

Col A	Results from NOW function	Results from Mo
Current date and time	13/03/2026 11:59	3
Date and Time 30 days ago	11/02/2026 11:59	2

ıay have to change the format of the cell

onth function

YEAR

What does it do?

This function returns the year month in a date in a number format. Do note that at times, you may have t
numeric to get the date in numeric format

Syntax

=YEAR(SERIAL_NUMBER)

Serial Number

Date from which you want to extract the year part

Example

Identify today's date and time

Col A	Result	Formula
29/03/2017 12:11	2017	=YEAR(A16)
22/02/1951	1951	=YEAR(A17)

Exercise 1

Use NOW() to calculate values in COL A and then use YEAR function to extract year

Col A	Results from NOW function	Results from Y
Current date and time	13/03/2026 11:59	202
Date and Time 365 days ago	13/03/2025 11:59	202

Exercise 2

Use YEAR, MONTH, DAY and DATE functions to get dates in COL A from Row 30 onwards. Row 29 has bee

Date	Results	
15/01/2016	Use this date to calculate dates from Row 30 onwards	
15/02/2016	15/02/2016	=DATE(YEAR(A28
15/03/2016	15/04/2016	
15/04/2016	15/05/2016	
15/05/2016	15/06/2016	
15/06/2016	15/07/2016	
15/07/2016	15/08/2016	
20/08/2017	20/09/2017	

o change the format of the cell to

ear function
6
5

n done as an example

3),MONTH(A28)+1,DAY(A28))

WEEKDAY

What does it do?

This function returns a number between 1 and 7 depending upon the day of the week. Default return is -
Saturday which will be 7. Do note that at times, you may have to change the format of the cell to numeric

Syntax

=WEEKDAY(SERIAL_NUMBER, [return_type])

Serial Number

Date from which you want to extract the year part

return_type

Default settings i.e. if you do not mention this, Sunday would be 1, Monday 2 and so on. If you select it as

Example

Identify today's weekday number basis different return_type value

Col A	Result	Formula
13/03/2026	6	=WEEKDAY(A20)
13/03/2026	5	=WEEKDAY(A21,2)
13/03/2026	4	=WEEKDAY(A22,3)
13/03/2026	2	=WEEKDAY(A23,14)

Exercise 1

Use WEEKDAY() to calculate weekdays of dates mentioned in 'Date' column using 'Return Value'

Dates	Return Value	Results
01/01/2017	1	01/01/1900 0:00
01/01/2017	2	01/01/1900 0:00
01/01/2017	3	01/01/1900 0:00

Exercise 2

If any of the dates in the column below falls on Tuesday, then change that date to Friday. Use Weekday fu
have to use IF function. 1st has been done as an example

Dates	Weekday	New Date
28/03/2017	3	24/03/2017
29/03/2017	4	29/03/2017
21/03/2017	3	24/03/2017
01/01/2016	6	01/01/2016
01/01/2017	1	01/01/2017
01/01/2015	5	01/01/2015
01/01/2014	4	01/01/2014
01/01/2013	3	04/01/2013

Sunday would be 1, Monday 2...so on till
: to get the date in numeric format

; 2, Monday will be given 1, Tuesday 2 and so on.

ıcntion to check weekdays with return value 1. You will

MINUTE

What does it do?

This function returns the minutes between 0 and 59

Syntax

=MINUTE(SERIAL_NUMBER)

Serial Number

Date or time from which you want to extract minutes

Example

Identify today's weekday number basis different return_type value

Col A	Result	Formula
13/03/2026 11:59	59	=MINUTE(A15)
10:15 am	15	=MINUTE(A16)
10:20 pm	20	=MINUTE(A17)

Exercise

Use MINUTE() to find out minutes from Col A

Col A	Results
31/01/2017 0:50	50
01/01/2017 0:01	1
12:45 am	45

VLOOKUP

What does it do?

This function is used to lookup a particular value in a table and then return data from a specific column

Syntax

=VLOOKUP(LookupValue, Table_Array, Column_Index_Num, [Range_Lookup])

LookupValue

The value which needs to be looked for in a particular table

Table_Array

The table in which we are checking for this value to be present and from which we want to retrieve data
you fix the range. So when you copy formula from one cell to another, the range does not update auton

Column_Index_Num

If the value is present in the table, then which column consists of the value we want the function to retu

Range_Lookup

It is optional - it has 2 values - TRUE for approximate match and FALSE (which is default value) for exact

Example 1

Below in the **Lookup** table, we have cost for different parts for different car makers. Use VLOOKUP() to

Lookup Table	Mercedes	BMW	Audi
GearBox	$2,500	$2,000	$3,000
Engine	$5,000	$4,999	$5,200
Steering	$1,250	$1,302	$1,100
Ignition	$250	$300	$200
CYHead	$300	$290	$310

Maker	Spare	Cost	Formula
Mercedes	Ignition	£250	=VLOOKUP(B31,$A$24:$B$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B32,$A$24:$C$28,3,FALSE)
Audi	Engine	£5,200	=VLOOKUP(B33,$A$24:$D$28,4,FALSE)
Audi	Steering	£1,100	=VLOOKUP(B34,$A$24:$D$28,4,FALSE)
Mercedes	Ignition	£250	=VLOOKUP(B35,$A$24:$B$28,2,FALSE)
Mercedes	CYHead	£300	=VLOOKUP(B36,$A$24:$B$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B37,$A$24:$C$28,3,FALSE)
BMW	Engine	£4,999	=VLOOKUP(B38,$A$24:$C$28,3,FALSE)

Example 2

Below are 3 tables, the first one consists of costs of 4 Items, the second one consists of its sales price. U
3rd table and then use Units and the difference between Sales Price and Cost to calculate profit or loss

Table 1	Table 2

Item Name	Cost
Pen	$5.10
Pencil	$1.10
Notepad	$1.50
Ink	$2.50

Item Name	Sales Price
Pen	$7.50
Pencil	$1.15
Notepad	$2.50
Ink	$3.75

Table 3					
Item Name	Units	Cost	Sales Price	Total Profit	For
Pen	500	$5.10	$7.50	$1,200.00	=VLOOKUP(A5
Pencil	100	$1.10	$1.15	$5.00	=VLOOKUP(A5
Notepad	250	$1.50	$2.50	$250.00	=VLOOKUP(A5
Ink	100	$2.50	$3.75	$125.00	=VLOOKUP(A5
Eraser	990	#N/A	#N/A		=VLOOKUP(A5

Total Profit = (Sales Price-Cost)*Units

Since Eraser was not present in Table 1 and in Table 2, excel returns an error #N/A implying that the data we are searching for is not present

Exercise 1

Below are 3 different tables, in the first table we have employees and their designation, in the second ta
we have their names and department. Use Vlookup() to fill all these details in the 4th table

Table 1	
E Name	Designation
Ben	AVP
Ken	Analayst
Rachel	AVP
Katie	Analyst
Helen	Analyst

Table 2	
E Name	Salary
Ben	$85,000
Ken	$60,000
Rachel	$65,000
Katie	$50,000
Helen	$45,000

Table 4			
E Name	Designation	Salary	Dept
Katie	Analyst	$50,000	HR
Helen	Analyst	$45,000	IT
Rachel	AVP	$65,000	IT
Ben	AVP	$85,000	Analytics
Richard	#N/A	#N/A	#N/A
Ken	Analayst	$60,000	Analytics

Exercise 2

Below in the **Lookup** table, we have cost for different parts for different car makers. Use VLOOKUP() to

Lookup Table	**GearBox**	**Engine**	**Steering**	**Ignition**	**CYHead**
Mercedes	$2,500	$5,000	$1,250	$250	$300
BMW	$2,000	$4,999	$1,302	$300	$290
Audi	$3,000	$5,200	$1,100	$200	$310

Maker	Mercedes	BMW	Audi	Audi	Mercedes
Spare	CYHead	GearBox	Gearbox	Steering	Ignition
Costs	#N/A				

corresponding to that value

ɜ. By pressing F2 key when you have selected your table,
natically

ɹrn

match of the value we are searching for in a table

get costs infront of items in the second table

1) The first parameter relates to spare part for which we want to get the costs
2) The second part refers to the data range - so Mercedes spare part information is availa
ofthe 'LookUp table' - so that is why you select only first 2 columns - 1st one consists of tl
which we are searching for and the second one consists of its cost. For BMW that informa
3rd column, thus our data range consists of 3 columns for the second paramter
3) The 3rd parameter refers to from which column we want to return the value, for Merc
2nd column of the selected range, for BMW it would be 3rd and for Audi it would be 4th
4) FALSE refers to us wanting an exact match and not an approximate one

se VLOOKUP function to put the cost and sales price in

mula for Cost	Formula for Sales Price
54,$A$46:$B$49,2,FALSE)	=VLOOKUP(A54,$D$46:$E$49,2,FALSE)
55,$A$46:$B$49,2,FALSE)	=VLOOKUP(A55,$D$46:$E$49,2,FALSE)
56,$A$46:$B$49,2,FALSE)	=VLOOKUP(A56,$D$46:$E$49,2,FALSE)
57,$A$46:$B$49,2,FALSE)	=VLOOKUP(A57,$D$46:$E$49,2,FALSE)
58,$A$46:$B$49,2,FALSE)	=VLOOKUP(A58,$D$46:$E$49,2,FALSE)

able we have their names and salary and in the 3rd one

Table 3	
E Name	**Dept**
Ben	Analytics
Ken	Analytics
Rachel	IT
Katie	HR
Helen	IT

get costs of items in the second table

Mercedes	BMW	BMW
Engine	GearBox	Engine

ıble in second column
he spare part name
ation is available in

edes that would be

HLOOKUP

What does it do?

This function is similar to VLOOKUP. In a VLOOKUP, we look at columns, whereas in HLOOKUP, we look a
a value in the first row of a table, and returns a value from the same column in that table.

Syntax

=HLOOKUP(LookupValue, Table_Array, Row_Index_Num, [Range_Lookup])

LookupValue

The value which needs to be looked for in a particular table

Table_Array

The table in which we are checking for this value to be present and from which we want to retrieve data
you fix the range. So when you copy formula from one cell to another, the range does not update auton

Row_Index_Num

If the value is present in the table, then which row consists of the value we want the function to return

Range_Lookup

It is optional - it has 2 values - TRUE for approximate match (which is default) and FALSE for exact match

Example

Below in the first table, we have sales commission for different cars. Use VLOOKUP() to get amount eacl

Car Type	Mercedes	BMW	Audi	VW	Ferrari
Commission	$500	$450	$550	$200	$1,000

Sold By	Maker	Commission	Formula
Ben	Mercedes	£500	=HLOOKUP(B29,$B$25:$F$26,2,FALSE)
Ben	BMW	£450	=HLOOKUP(B30,$B$25:$F$26,2,FALSE)
Alex	Audi	£550	=HLOOKUP(B31,$B$25:$F$26,2,FALSE)
Gary	Audi	£550	=HLOOKUP(B32,$B$25:$F$26,2,FALSE)
Helen	Ferrari	£1,000	=HLOOKUP(B33,$B$25:$F$26,2,FALSE)
Helen	Ferrari	£1,000	=HLOOKUP(B34,$B$25:$F$26,2,FALSE)
Ken	BMW	£450	=HLOOKUP(B35,$B$25:$F$26,2,FALSE)
Arnold	Mercedes	£500	=HLOOKUP(B36,$B$25:$F$26,2,FALSE)

Exercise

Below in the **Lookup** table, we have cost for different parts for different car makers. Use HLOOKUP() to
been done as an example

Lookup Table	Mercedes	BMW	Audi
GearBox	$2,500	$2,000	$3,000
Engine	$5,000	$4,999	$5,200
Steering	$1,250	$1,302	$1,100

Ignition	$250	$300	$200
CYHead	$300	$290	$310

Maker	Mercedes	BMW	Audi	Audi	Mercedes
Spare	CYHead	GearBox	Gearbox	Steering	Ignition
Costs	£300				

=HLOOKUP(B50,$B$42:$B$47,6,FALSE)

at rows. The HLOOKUP function looks for

a. By pressing F2 key when you have selected your table,
natically

n of the value we are searching for in a table

h employee will get based on car they sold

1) The first parameter relates to car for which we want to get the commission amount
2) The second part refers to the data range - so car names are in the 1st row of table 1 an information is in 2nd row of table 1
3) The 3rd parameter refers to from which row we want to return the value - in this case which consists of the commission amount
4) FALSE refers to us wanting an exact match and not an approximate one

get costs of items in the second table. First one has

Mercedes	BMW	BMW
Engine	GearBox	Engine

d the commission

it is the 2nd row

TRANSPOSE

What does it do?

The TRANSPOSE function copies data from vertical cells to horizontal or vice versa

Syntax

=TRANSPOSE(Range)

Range

Data which you want to change from vertical to horizontal or vice-versa

How to use Transpose function

The transpose range must be the same size as the original range.
The function needs to be entered as an array formula.
To enter an array formula you must first highlight all the cells where the formula is required.
Next type the formula, such as =TRANSPOSE(A1:A5).
Finally press Ctrl+Shift+Enter to confirm it.
If changes need to be made to the formula, the entire array has to be highlighted, the edits can then be
The curly braces " { } " surrounding the function indicate that it is an array formula

Example

Use TRANSPOSE function to change the cells from vertical to horizontal

Employee ID	Jan	Feb
E001	10	30
E002	40	50
E003	70	80

EID	E001	E002	E003
Jan	10	40	70
Formula EID	=TRANSPOSE(A24:A26)		
Formula Jan	=TRANSPOSE(B24:B26)		

Exercise

Use TRANSPOSE function to change the cells from vertical to horizontal for Employee ID and Feb using s

Employee ID	Jan	Feb
E001	10	30
E002	40	50
E003	70	80

EID	E001	E002	E003
Jan	10	40	70

made and the Ctrl+Shift+Enter used to confirm it.

teps mentioned from Row 12 onwards

MATCH

What does it do?

This function is used to locate the position of a lookup value in a row, column, or table.

Syntax

=MATCH (Lookupvalue, Lookup_Array, [Match_type])

LookupValue

The value which needs to be looked for in a particular column, row or table

Lookup_Array

The column, table or table in which we are checking for this value to be present

Match_Type

It is optional - it has 3 values --> -1 gives the smallest value position that is greater than or equal to valu search and 1 gives the largest value that is leass than or equal to the value we are searching for. Default

Example 1

Below in the **Lookup** table, we have cost for different parts for different car makers. Use VLOOKUP() to

Lookup Table	Mercedes	BMW	Audi
GearBox	$2,500	$2,000	$3,000
Engine	$5,000	$4,999	$5,200
Steering	$1,250	$1,302	$1,100
Ignition	$250	$300	$200
CYHead	$300	$290	$310

Maker	Spare	Cost	Formula
Mercedes	Ignition	£250	=VLOOKUP(B31,$A$24:$B$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B32,$A$24:$C$28,3,FALSE)
Audi	Engine	£5,200	=VLOOKUP(B33,$A$24:$D$28,4,FALSE)
Audi	Steering	£1,100	=VLOOKUP(B34,$A$24:$D$28,4,FALSE)
Mercedes	Ignition	£250	=VLOOKUP(B35,$A$24:$B$28,2,FALSE)
Mercedes	CYHead	£300	=VLOOKUP(B36,$A$24:$B$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B37,$A$24:$C$28,3,FALSE)
BMW	Engine	£4,999	=VLOOKUP(B38,$A$24:$C$28,3,FALSE)

Example 2

Below are 3 tables, the first one consists of costs of 4 Items, the second one consists of its sales price. U 3rd table and then use Units and the difference between Sales Price and Cost to calculate profit or loss

Table 1	
Item Name	**Cost**
Pen	$5.10
Pencil	$1.10

Table 2	
Item Name	**Sales Price**
Pen	$7.50
Pencil	$1.15

Notepad	$1.50
Ink	$2.50

Notepad	$2.50
Ink	$3.75

Table 3					
Item Name	**Units**	**Cost**	**Sales Price**	**Total Profit**	**For**
Pen	500	$5.10	$7.50	$1,200.00	=VLOOKUP(A5
Pencil	100	$1.10	$1.15	$5.00	=VLOOKUP(A5
Notepad	250	$1.50	$2.50	$250.00	=VLOOKUP(A5
Ink	100	$2.50	$3.75	$125.00	=VLOOKUP(A5
Eraser	990	#N/A	#N/A		=VLOOKUP(A5

Total Profit = (Sales Price-Cost)*Units

Since Eraser was not present in Table 1 and in Table 2, excel returns an error #N/A implying that the data we are searching for is not present

Exercise 1

Below are 3 different tables, in the first table we have employees and their designation, in the second ta
we have their names and department. Use Vlookup() to fill all these details in the 4th table

Table 1	
E Name	**Designation**
Ben	AVP
Ken	Analayst
Rachel	AVP
Katie	Analyst
Helen	Analyst

Table 2	
E Name	**Salary**
Ben	$85,000
Ken	$60,000
Rachel	$65,000
Katie	$50,000
Helen	$45,000

Table 4			
E Name	**Designation**	**Salary**	**Dept**
Katie	Analyst	$50,000	HR
Helen	Analyst	$45,000	IT
Rachel	AVP	$65,000	IT
Ben	AVP	$85,000	Analytics
Richard	#N/A	#N/A	#N/A
Ken	Analayst	$60,000	Analytics

Exercise 2

Below in the **Lookup** table, we have cost for different parts for different car makers. Use VLOOKUP() to

Lookup Table	GearBox	Engine	Steering	Ignition	CYHead
Mercedes	$2,500	$5,000	$1,250	$250	$300

BMW	$2,000	$4,999	$1,302	$300	$290
Audi	$3,000	$5,200	$1,100	$200	$310

Maker	Mercedes	BMW	Audi	Audi	Mercedes
Spare	CYHead	GearBox	Gearbox	Steering	Ignition
Costs					

e we are searching, 0 gives the value for the exact
is 1

get costs infront of items in the second table

1) The first parameter relates to spare part for which we want to get the costs
2) The second part refers to the data range - so Mercedes spare part information is availa
ofthe 'LookUp table' - so that is why you select only first 2 columns - 1st one consists of tl
which we are searching for and the second one consists of its cost. For BMW that informa
3rd column, thus our data range consists of 3 columns for the second paramter
3) The 3rd parameter refers to from which column we want to return the value, for Merc
2nd column of the selected range, for BMW it would be 3rd and for Audi it would be 4th
4) FALSE refers to us wanting an exact match and not an approximate one

se VLOOKUP function to put the cost and sales price in

mula for Cost	Formula for Sales Price
54,$A$46:$B$49,2,FALSE)	=VLOOKUP(A54,$D$46:$E$49,2,FALSE)
55,$A$46:$B$49,2,FALSE)	=VLOOKUP(A55,$D$46:$E$49,2,FALSE)
56,$A$46:$B$49,2,FALSE)	=VLOOKUP(A56,$D$46:$E$49,2,FALSE)
57,$A$46:$B$49,2,FALSE)	=VLOOKUP(A57,$D$46:$E$49,2,FALSE)
58,$A$46:$B$49,2,FALSE)	=VLOOKUP(A58,$D$46:$E$49,2,FALSE)

able we have their names and salary and in the 3rd one

Table 3	
E Name	**Dept**
Ben	Analytics
Ken	Analytics
Rachel	IT
Katie	HR
Helen	IT

get costs of items in the second table

Mercedes	BMW	BMW
Engine	GearBox	Engine

ıble in second column
he spare part name
ation is available in

edes that would be

SUM

What does it do?

This function returns the sum of the cells or range selected

Syntax

=SUM(Number 1, Number 2...)

Number 1

The first value to sum

Number 2 and so on

The second (and so on..) value to sum

Example 1

Use SUM function to get the sum of cells

Col A	Col B	Col C	Result	Formula
5	12	17	34	=SUM(A18:C18)
10	18	23	51	=SUM(A19,B19,C19)
20	100	999	1119	=A20+B20+C20

Example 2

Use SUM function to get the sum of cells

Row A	5	9	12	21
Row B	1	56	345	23
Row C	2	6	23	34
Row D	4	12	547	12
Total	12	83	927	90
Formula	=SUM(B24:B27)	=C24+C25+C26+C27	=SUM(D24,D25,D26,D27)	=SUM(E24:E25,E26:E27)

Example 3

Get the SUM of all cells below

Values to be summed			Total	Formula
5		3	16	=SUM(A34,A36,C34,C36,B35:B36)
	1			
1	2	4		

Note

If you want to sum the entire column I suppose, all you need to do is =SUM(I:I), it will select the entire colu
Similarly, you can select the entire row as well. If you write, =SUM(41:42), It will return the sum of all numb

Exercise 1

Use SUM function to calculate sum of the following values

Values to be summed			Result
1			101
100		500	
	200		

Exercise 2

Use IF and SUM function to calculate sum of the following values if conditions meet. If the conditions do n
example

Condition to meet	Col B	Col C	Col D	Col E	Result
Col B > 1	5	1	2	4	12
Col C > 50	9	56	6	12	83
Col C > 500	12	345	23	547	927
Col E > 80	84.3	59.8	80.9	80.5	305.5
Col D > 35	9.8	0.2	38	80.9	128.9

Exercise 3

Use IF, AND and SUM functions to calculate sum of the following values if conditions meet. If the condition
as an example

First Condition	Second Condition	Col B	Col C	Col D	Col E
Col B > 1	**Col D >= 2**	5	1	2	4
Col C > 50	**Col A > 5**	9	56	6	12
Col C > 500	**Col E > 500**	12	345	23	547
Col E > 80	**Col A > 80**	84.3	59.8	80.9	80.5
Col D > 35	**Col B >= 9.8**	9.8	0.2	38	80.9

Exercise 4

Use IF, OR and SUM functions to calculate sum of the following values if either of the 2 conditions meet. If
has been done as an example

First Condition	Second Condition	Col B	Col C	Col D	Col E
Col B > 1	**Col D >= 2**	5	1	2	4
Col C > 50	**Col A > 5**	9	56	6	12
Col C > 500	**Col E > 500**	12	345	23	547
Col E > 80	**Col A > 80**	84.3	59.8	80.9	80.5
Col D > 35	**Col B >= 9.8**	9.8	0.2	38	80.9

ımn I and give you the sum of the numbers in it.
ɔers in Rows 41 and 42.

ot meet, then return value 0. First has been done as an

=IF(B54>1,SUM(B54:E54),0)

ıs do not meet, then return value 0. First has been done

Result
12
83
927
305.5
128.9

=IF(AND(C65>1,E65>=2),SUM(C65:F65),0)

the conditions do not meet, then return value 0. First

Result
12
83
927
305.5
128.9

=IF(OR(C76>1,E76>=2),SUM(C76:F76),0)

SUMIF

What does it do?

This function returns the sum of the cells which meet a given criteria

Syntax

=SUMIF(Range, Criteria, [sum_range])

Range

Range of cells which we need the criteria to evaluate

Criteria

This refers to the condition which we want to check, it can be text, number, dates

Sum Range

This is optional, if not present, excel will consider numbers in the 'Range'. Otherwise, it specifies the

Example 1

Use SUMIF function to calculate salary of all employees in Analytics and then for Sales

Employee ID	Department	Designation	Date of Joining	Salary
E001	HR	AVP	01/01/2017	$40,000
E002	Analytics	AVP	01/12/2016	$75,000
E003	Sales	AVP	01/12/2016	$30,000
E004	Analytics	AVP	01/11/2016	$80,000
E005	HR	Analyst	01/10/2016	$35,000
E006	Sales	Analyst	01/11/2016	$25,000
E007	Analytics	Analyst	01/10/2016	$60,000
E008	Sales	Analyst	01/11/2016	$20,000
E009	IT	Analyst	01/10/2016	$40,000
E010	IT	Analyst	01/10/2016	$38,000

	Total	Formula
Salary for Analytics Group	$215,000	=SUMIF(B22:B31,"Analytics",E22:E31)
Salary for Analytics Group	$78,000	=SUMIF(B22:B31,"IT",E22:E31)

Example 2

Use SUMIF function to calculate salary of all employees who joined before 1st Dec 2016

Employee ID	Department	Designation	Date of Joining	Salary
E001	HR	AVP	01/01/2017	$40,000
E002	Analytics	AVP	01/12/2016	$75,000
E003	Sales	AVP	01/12/2016	$30,000
E004	Analytics	AVP	01/11/2016	$80,000

E005	HR	Analyst	01/10/2016	$35,000
E006	Sales	Analyst	01/11/2016	$25,000
E007	Analytics	Analyst	01/10/2016	$60,000
E008	Sales	Analyst	01/11/2016	$20,000
E009	IT	Analyst	01/10/2016	$40,000
E010	IT	Analyst	01/10/2016	$38,000

Salary for employees who joined before 1st Dec'16	Total	Formula
	$298,000	=SUMIF(D41:D50,"<01-Dec-2016",E41:E50)

Exercise 1

Use SUMIF function to calculate salary of all employees who are AVP

Employee ID	Department	Designation	Date of Joining	Salary
E001	HR	AVP	01/01/2017	$40,000
E002	Analytics	AVP	01/12/2016	$75,000
E003	Sales	AVP	01/12/2016	$30,000
E004	Analytics	AVP	01/11/2016	$80,000
E005	HR	Analyst	01/10/2016	$35,000
E006	Sales	Analyst	01/11/2016	$25,000
E007	Analytics	Analyst	01/10/2016	$60,000
E008	Sales	Analyst	01/11/2016	$20,000
E009	IT	Analyst	01/10/2016	$40,000
E010	IT	Analyst	01/10/2016	$38,000

	Result	Formula
Salary for Analytics Group	$225,000	$ 675,000.00

Exercise 2

Use SUMIF function to calculate salary of all employees joined on 1st Oct 2016

Employee ID	Department	Designation	Date of Joining	Salary
E001	HR	AVP	01/01/2017	$40,000
E002	Analytics	AVP	01/12/2016	$75,000
E003	Sales	AVP	01/12/2016	$30,000
E004	Analytics	AVP	01/11/2016	$80,000
E005	HR	Analyst	01/10/2016	$35,000
E006	Sales	Analyst	01/11/2016	$25,000
E007	Analytics	Analyst	01/10/2016	$60,000
E008	Sales	Analyst	01/11/2016	$20,000
E009	IT	Analyst	01/10/2016	$40,000
E010	IT	Analyst	01/10/2016	$38,000

	Result	Formula
Salary for Analytics Group	$173,000	$ 225,000.00

cells which need to be added if the condition holds

SUMPRODUCT

What does it do?

This function returns the sum after multiplying numbers in an array. It can also be used to count cells ba

Syntax

=SUMPRODUCT(Array1, [Array2], [Array3],…and so on)

Array1

Range of cells which first need to be multiplied and then summed

Array2

It is optional - the second array or range to multiply and then summed

Example 1

Use SUMPRODUCT function to calculate total cost faced by the superstore

Items	Perishable?	Cost	No. of Units
I001	NO	£1.50	100
I002	YES	£1.99	150
I003	YES	£2.05	200
I004	NO	£3.50	50
I005	YES	£4.90	400
I006	NO	£5.00	1,000

	Total	Formula
Total Cost	$7,994	=SUMPRODUCT(C18:C23,D18:D23)

Example 2

Use SUMPRODUCT function to count total number of Perishable items

Items	Perishable?	Total Cost
I001	NO	£150
I002	YES	£299
I003	YES	£410
I004	NO	£175
I005	YES	£1,960
I006	NO	£5,000

	Total	Formula
Total Items	3	=SUMPRODUCT(--(B32:B37="YES"))
Cost for Perishable Items	$2,669	=SUMPRODUCT(--(B32:B37="YES"),C32:C37)

Exercise 1

Use SUMPRODUCT function to calculate total cost faced by the superstore

Items	Perishable?	Sales	No. of Units
I001	NO	£3.10	500
I002	YES	£2.50	150
I003	YES	£10.00	100
I004	NO	£1.00	500
I005	YES	£4.00	200
I006	NO	£2.00	2,000

	Total	Formula
Total Sales	$8,225	3

Exercise 2

Use SUMPRODUCT function to count total number of non Perishable items

Items	Perishable?	Total Cost
I001	NO	£150
I002	YES	£299
I003	YES	£410
I004	NO	£175
I005	YES	£1,960
I006	NO	£5,000

	Total
Total Items	0
Cost Non Perishable Items	3

ased on criteria provided

RAND

What does it do?

This function returns random numbers greater than 0 and less than 1. The numbers change on recalcula

Syntax

=RAND()

No Paramter

No Parameter is required for this function

Example

Use RAND() to generate some random numbers

Random No.s	Formula
0.865705632	=RAND()
0.181624553	=RAND()
0.797740136	=RAND()
0.46247027	=RAND()
0.91623168	=RAND()
0.54992059	=RAND()

ıtion

RANDBETWEEN

What does it do?

This function returns random numbers between specified numbers

Syntax

=RANDBETWEEN(BOTTOM, TOP)

BOTTOM

Lower Limit

TOP

Upper Limit

Example

Use RAND() to generate some random numbers

Random No.s	Formula
26736	=RANDBETWEEN(0,100000)
19881	=RANDBETWEEN(0,100000)
26276	=RANDBETWEEN(0,100000)
2916	=RANDBETWEEN(0,5000)
421	=RANDBETWEEN(0,5000)
4419	=RANDBETWEEN(0,5000)

ROUND

What does it do?

This function rounds the numbers to specified digits.

Syntax

=ROUND(Number,num_digits)

Number

Number which you want to round up

num_digits

Number of decimal places to which you want it to round up

Example

Use ROUND function reduce decimal places of the numbers

Numbers	Rounded	Formula
45.3976106	45.40	=ROUND(A18,2)
50.6546265	50.70	=ROUND(A19,1)
85.4139093	85.414	=ROUND(A20,3)
33.5893471	34	=ROUND(A21,0)

Exercise

Use ROUND function reduce decimal places of the numbers

Items	Result	Formula
81.373	81.37	81.370
7.9597	8.00	7.960
521.2578	521.26	521.260
4.699	4.70	4.700

EVEN

What does it do?

This function rounds a positive number up and negative number down to the nearest even integer

Syntax

=EVEN(Number/Cell)

Text

Cell which you want to convert to the nearest even integer

Example

Use EVEN function on the following examples

Numbers	Rounded	Formula
45.3976106	46	=EVEN(A15)
-50.6546265	-52	=EVEN(A16)
85.4139093	86	=EVEN(A17)
31.5893471	32	=EVEN(A18)

Exercise

Use EVEN function and round up numbers to nearest even number

Items	Result
1111.111	1,112.00
95.3	96.00
51.5	52.00
-4.699	-6.00

ODD

What does it do?

This function rounds a positive number up and negative number down to the nearest odd integer

Syntax

=ODD(Number/Cell)

Text

Cell which you want to convert to the nearest odd integer

Example

Use ODD function on the following examples

Numbers	Rounded	Formula
45.3976106	47	=ODD(A15)
-50.6546265	-51	=ODD(A16)
85.4139093	87	=ODD(A17)
33.5893471	35	=ODD(A18)

Exercise

Use ODD function and round up numbers to nearest odd number

Items	Result
1111.111	1,113.00
95.3	97.00
51.5	53.00
-4.699	-5.00

MOD

What does it do?

This function returns the remainder after a number is divided by a divisor

Syntax

=MOD(number,divisor)

Number

This is the number which you want to divide

Divisor

The number to divide with

Example

Use MOD function on the following examples

Number	Divisor	Remainder	Formula
3	5	3	=MOD(A18,B18)
19	3	1	=MOD(A19,B19)
8	2	0	=MOD(A20,B20)
-8.5	3	0.5	=MOD(A21,B21)

Exercise

Use MOD function to find the remainders

Number 1	Number 2	Result
5	6	5
9	2	1
8	5	3
-4.699	-1	-0.6985

MIN

What does it do?

This function returns the smallest number in a given set of values. It ignores logical values and text

Syntax

=MIN(Number1,Number2 and so on..)

Number1,Number2 and so on..

List of values from which you want to find the mimimum value

Example

Use MIN function to find the smallest value of the 3 numbers present for each row

Number1	Number2	Number3	Result	Formula
3	5	3	3	=MIN(A15:C15)
19	a	1	1	=MIN(A16:C16)
8	b	c	8	=MIN(A17:C17)
-8.5	3	0.5	-8.5	=MIN(A18:C18)

Exercise

Use MIN and IF function on the following and if the value is less than 5, then flag it as 'Less than 5' else '

Number1	Number2	Number3	Result
3	5	3	Less than 5
19	a	1	Less than 5
8	b	c	eater than equal to 5
-8.5	3	0.5	Less than 5

Greater than equal to 5'

MAX

What does it do?

This function returns the largest number in a given set of values. It ignores logical values and text

Syntax

=MAX(Number1,Number2 and so on..)

Number1,Number2 and so on..

List of values from which you want to find the maximum value

Example

Use MAX function to find the smallest value of the 3 numbers present for each row

Number1	Number2	Number3	Result	Formula
3	5	3	5	=MAX(A15:C15)
19	a	1	19	=MAX(A16:C16)
8	b	c	8	=MAX(A17:C17)
-8.5	3	0.5	3	=MAX(A18:C18)

Exercise

Use MAX and IF function on the following and if the value is less than 5, then flag it as 'Less than 5' else

Number1	Number2	Number3	Result
3	5	3	eater than equal to 5
19	a	1	eater than equal to 5
8	b	c	eater than equal to 5
-8.5	3	0.5	Less than 5

'Greater than equal to 5'

AVERAGE

What does it do?

This function returns the average value of the given numbers - It ignores text values and blank cells

Syntax

=AVERAGE(Number1,Number2 and so on..)

Number1,Number2 and so on..

List of values from which you want to find the average value

Example

Use AVERAGE function to find the smallest value of the 3 numbers present for each row

Number1	Number2	Number3	Result	Formula
3		5	4	=MAX(A15:C15)
19	a	1	10	=MAX(A16:C16)
8	b	c	8	=MAX(A17:C17)
-8.5	3	0.5	-1.666666667	=MAX(A18:C18)

Exercise

Calculate AVERAGE for the following numbers

Number1	Number2	Number3	Result
3	5	3	3.666666667
19	a	1	10
8	b	c	8
1.5		4.5	3

SMALL

What does it do?

This function returns the nth smallest number from the given set of values

Syntax

=SMALL(Array, nth small number from the list)

Array

These are list of numbers from which nth smallest number needs to be found out

nth smallest number

The nth smallest number which needs to be found out like 5th smallest number etc

Example

Use SMALL function to find the nth smallest value from the set of values in the range A18:C21

Array/List of Numbers			nth smallest number	Result	Form
3	1	5	1	-8.5	=SMALL(A18
19	50	1	5	3	=SMALL(A18
8	12	20	4	1	=SMALL(A18
-8.5	3	0.5	12	50	=SMALL(A18

Exercise

Calculate nth smallest number for the following numbers

Array/List of Numbers			nth smallest number	Result
3	5	3	1	3
19	2	1	2	2
8	5	3	3	8
1.5	1	4.5	4	#NUM!

ula
3:C21,D18)
3:C21,D19)
3:C21,D20)
3:C21,D21)

LARGE

What does it do?

This function returns the nth largest number from the given set of values

Syntax

=LARGE(Array, nth largest number from the list)

Array

These are list of numbers from which nth largest number needs to be found out

nth largest number

The nth largest number which needs to be found out like 5th largest number etc

Example

Use SMALL function to find the nth smallest value from the set of values in the range A18:C21

Array/List of Numbers			nth largest number	Result	Form
3	1	5	1	50	=LARGE(A18
19	50	1	5	8	=LARGE(A18
8	12	20	4	12	=LARGE(A18
-8.5	3	0.5	12	-8.5	=LARGE(A18

Exercise

Calculate nth largest number for the following numbers

Array/List of Numbers			nth largest number	Result
3	5	3	1	5
19	2	1	2	2
8	5	3	3	3
1.5	1	4.5	4	#NUM!

ula
:C21,D18)
:C21,D19)
:C21,D20)
:C21,D21)

COUNT

What does it do?

This function returns the count of number of cells which consist of numbers

Syntax

=COUNT([value1],[value[2], and so on..)

value1

First cell in the range

value 2 and so on

Second, third and so on cells in the range

Example

Use COUNT function to find the number of numeric values

List of Numbers			Result	Formula
3	1	5	3	=COUNT(A18:C18)
19	A	1	2	=COUNT(A19:C19)
8		20	2	=COUNT(A20:C20)
-8.5		B	1	=COUNT(A21:C21)

Exercise 1

Use COUNT function to find the number of numeric values

List of Numbers			Result
3		3	2
b	2	1	2
c	5	3	2
1.5	1	4.5	3

Exercise 2

Use IF function to print 'Y' if count of numbers is >=2 else print 'N'

List of Numbers			Result
3		3	Y
b	2	1	Y
c	5	3	Y
1.5		D	N

COUNTIF

What does it do?

This function returns the count of number of cells which consist of numbers and meet a given condition

Syntax

=COUNTIF(Range,Criteria)

Range

Range of cells which you want to count

Criteria

Condition which you want to check and then count the cells on the basis of it

Example

Use COUNTIF function on Col A, B, C and D to find the number of numeric values, based on values being

Col A	Col B	Col C	Col D	Col E	Result
3	1	5	3	Values > 1	3
19	A		0	Values = 0	1
8	D	20	2	Values >= 2	3
-8.5	0	B	-1	Values < 0	2

Exercise

Use COUNTIF function on Col A, B, C and D to find the number of numeric values, based on values being

Col A	Col B	Col C	Col D	Col E	Result
1	-1	5	3	Values > 2	2
19	A		0	Values >= 0	1
8	-50	20	2	Values >= 20	2
-8.5	0		-1	Values <= 0	0

; greater or less as specified in Col E

Formula
=COUNTIF(A18:D18,">1")
=COUNTIF(A19:D19,"=0")
=COUNTIF(A20:D20,">=2")
=COUNTIF(A21:D21,"<0")

; greater or less as specified in Col E

STDEV.S

What does it do?

This function gives the standard deviation of the given set of values or range of values. In terms of statis
better, as it implies that there is not much variance in the data. Blank cells, text, and error values are igr

Syntax

=STDEV.S([number1],[number2] and so on..)

[number1],[number2] and so on..

Numbers for which you want to find out standard deviation

Example

Calculate standard devation for the following set of numbers

Range of Numbers				STD DEV	Form
42	45	20	22	13.07351011	=STDEV.S(
27	10	34	50	16.58061117	=STDEV.S(
26	32	11	26	8.958236434	=STDEV.S(
42	40	47	47	3.559026084	=STDEV.S(

Exercise

Calculate standard devation for the following set of numbers

Range of Numbers				Result
7	7	2	3	2.62995564
1	2	1	6	2.380476143
5	8	8	6	1.5
5	8	7	4	1.825741858

stics, the lower the standard deviation the
nored.

ula
A16:D16)
A17:D17)
A18:D18)
A19:D19)

CORREL

What does it do?

This function returns the correlation coefficient between 2 datasets. Correlation lies between -1 and 1 c
correlation. For ex: as a baby grows older, the height or weight increases - implying that there is a relati
weight and with the increase in one the other is also increasing i.e. positive correlation

Syntax

=CORREL([Array1],[Array2])

[Array1],[Array2]

The 2 different arrays for which you want to calculate correlation

Example

Calculate correlation between Age and Weight for the following set of numbers

Age	Weight
1	5
2	14
3	15
4	20
5	24
6	30
7	35
8	38
9	40
10	45

Correlation	Formula
0.993023515	=CORREL(A17:A26,B17:B26)

Exercise

Calculate correlation between Age and Height for the following set of numbers

Age	Height
1	20
2	35
3	45
4	60
5	80
6	100
7	120
8	122
9	125
10	135

alled negative correlation and postive
on between the age and height or age and

MEDIAN

What does it do?

This function returns number which is in the middle of all numbers in a dataset

Syntax

=MEDIAN([number1],[number2], and so on..)

[number1],[number2], and so on..

The list of numbers for which you want to find the median

Example

Calculate median for the following numbers

Table A
89
91
65
59
60
86
51
95
84
88
92
85
74
76
60
65
93
98
64
77
89

Median	Formula
84	=MEDIAN(A15:A35)

If you sort the values ascending, and count the number of observations here, we have 21 (in Table A) and the middle observation would be 11th observation which has a value of 84. If the number of observations would have been 20 (as is the case in Table B), then median would have been 10th + 11th observation divided by 2 which

Table B
94
57
80
91
97
84
79
53
50
53
78
54
85
54
98
92
84
71
100
94

Median	Formula
82	=MEDIAN(G15:G34)

PMT, NPER, PV, FV

What does PMT do?

It calculates the payment of a loan based on constant number of payments and a fixed interest rate

Syntax

=PMT(rate,nper, pv, [fv], [type])

Rate
Interest Rate

Nper
Number of payments

PV
Present Value of loan taken

FV and Type
FV and Type are both optional arguments with FV standing for future value of the loan. Types tells whet
made at the end of the month or not

Suppose person ABC buys a car and agrees to a monthly payment based on an interest rate of 4% over 4
payments). The loan amount is $80000

Rate	0.33%	Since payments will be monthly, dividing interest rate by 12
NPER	48	
PV	$80,000	The PMT is negative i.e. highlighted red because it is im
PMT	$1,806	ABC will have to give this much amount of money ever
PV	**($1,806.32)**	

NPER	($1,765.67)
PV	($1,771.56)
FV	($1,806.90)

It calculates the number of payments required based on fixed interest rate, constant periodic payments

Syntax

=NPER(rate,pmt, pv, [fv], [type])

Rate
Interest Rate

PMT
Regular Payment amount

PV
Present Value of loan taken

FV and Type

FV and Type are both optional arguments with FV standing for future value of the loan. Types tells whet made at the end of the month or not

Rate	0.33%
PMT	$1,500
PV	$80,000
NPER	59
Formula	**=NPER(B50,B51,B52)**

ABC will have to make 59 motnhly payments of

What does PV do?

It tells the present worth of the future payments (present value)

Syntax

=PV(rate,nper, pmt, [fv], [type])

Rate

Interest Rate

NPER

Number of payments

PMT

Regular Payment amount

FV and Type

FV and Type are both optional arguments with FV standing for future value of the loan. Types tells whet made at the end of the month or not

Rate	0.33%
PMT	$1,806
NPER	48
PV	$79,986
Formula	**=PV(B76,B78,B77)**

Present Value of 48 future payments of $1,806 is

What does FV do?

It tells the future worth of constant payments at fixed interest rate (fixed value)

Syntax

=PV(rate,nper, pmt, [pv], [type])

Rate

Interest Rate

NPER
Number of payments

PMT
Regular Payment amount

PV and Type
PV and Type are both optional arguments with PV standing for present value of the loan. Types tells wh
made at the end of the month or not

Rate	0.33%
PMT	$1,806
NPER	48
FV	$0
Formula	**=FV(B102,B104,B103,B79)**

If FV is $0, it implies that we will be able to repay the lo
$80,000 we had initially taken by paying 48 monthly in
$1,806 each

her the payment will be

4 years (i.e. 48 monthly

nplying that
y month

her the payment will be

$1,500

her the payment will be

$79,986

ether the payment will be

oan of
stallments of